

Mark schemes

Q1.

(a) $y_A = 3(2^0 + 1)$

Substituting $x = 0$ PI

M1

$= 6$

A1

2

(b) $h = 2$

PI

B1

Integral = $h/2 \{ \dots \}$

$\{ \dots \} = f(0) + 2[f(2) + f(4)] + f(6)$

OE summing of areas of the three traps.

M1

$\{ \} = 6 + 2[3 \times 5 + 3 \times 17] + 3 \times 65 = 6 + 2[15 + 51] + 195$

Condone 1 numerical slip {ft on (a) for $f(0)$ if not recovered}

[Sum of 3 traps. = $21 + 66 + 246$]

A1

Integral = 333

CAO

A1

4

(c) (i) $21 = 3(2^x + 1) \Rightarrow 2^x = 6$

AG (be convinced)

B1

1

(ii) $\log_{10} 2^x = \log_{10} 6$

*Take ln or $\log_{10}$ of both sides of $a^x = b$
or other relevant base if clear. The
equation $a^x = b$ used must be correct.*

M1

$x \log_{10} 2 = \log_{10} 6$

Use of $\log 2^x = x \log 2$ OE

m1

$$x = \frac{\lg 6}{\lg 2} = 2.5849\dots = 2.58 \text{ to 3sf}$$

Both method marks must have been awarded.

A1

3

[10]

Q2.

(a) (i) $t = 0: x = 3$

B1

1

(ii) $t = 14: x = 15 - 12e^{-1}$

or $15 - 12e^{\frac{-14}{14}}$

$= 10.6$

CAO

M1

A1

2

(b) (i) $-5 = -12e^{-\frac{t}{14}}$

substitute $x = 10$; rearrange to form $p = qe^{-\frac{t}{14}}$

M1

$$\ln\left(\frac{5}{12}\right) = -\frac{t}{14} \quad (\text{OE})$$

take lns correctly

m1

$$t = 14 \ln\left(\frac{12}{5}\right)$$

must come from correct working

A1

3

(ii) $t = 12.256... \approx 12$ days

*ft on a, b if a > b; accept t = 12 NMS
 Accept 12 from incorrect working in (b)(i)
 Accept 13 if 12.2 or 12.3 seen*

B1F

1

(c) (i) $\frac{dx}{dt} = -\frac{1}{14}x - 12e^{-\frac{t}{14}}$

differentiate; allow sign error

condone $\frac{dy}{dx}$ used consistently

M1

$$= -\frac{1}{14}(x - 15)$$

Or $\frac{1}{14}\left(12e^{-\frac{t}{14}}\right)$ and $12e^{-\frac{t}{14}} = 15 - x$ seen

m1

$$= \frac{1}{14}(15 - x)$$

AG - be convinced CSO

A1

3

Alt: $t = -14 \ln\left(\frac{15 - x}{12}\right)$

attempt to solve given equation for t

(M1)

$$\frac{dt}{dx} = \frac{-14\left(-\frac{1}{12}\right)}{\left(\frac{15 - x}{12}\right)}$$

differentiate wrt x, with $\frac{1}{15 - x}$ seen; OE

(m1)

$$\frac{dt}{dx} = \frac{14}{15-x} \Rightarrow \frac{dx}{dt} = \frac{1}{14}(15-x)$$

AG – be convinced

(A1)

(3)

Alt: (backwards)

$$\int \frac{dx}{15-x} = \int \frac{dt}{14} = \pm 14 \ln(15-x) = t + c$$

(M1)

$$\text{Use (0,3): } -14 \ln(15-x) + 14 \ln 12 = t$$

(m1)

$$\text{Solve for } x: x = 15 - 12e^{-\frac{t}{14}}$$

All steps shown

(A1)

(3)

(ii) rate of growth = 0.5 (cm per day)

Accept $\frac{7}{14}$

B1

1

[11]

Q3.

(a) $\log 0.8^x = \log 0.05$ OR $x = \log_{0.8} 0.05$ **(M1)**

NMS: SC B2 for 13.425 or better
(B1 for 13.4 or 13.43; 13.42)

M1

$$x \log_{10} 0.8 = \log_{10} 0.05 \text{ oe}$$

A1

$x = 13.425$ to 3dp OR 13.425 **(A2)**
(else A1 for 1 or 2dp)

Condone greater accuracy

A1

3

(b) (i) $\frac{a}{1-r}$

$$S_{\infty} = \frac{a}{1-r} \text{ used}$$

M1

$$\frac{a}{1-r} = 5a \Rightarrow a = 5a(1-r)$$

Or better

A1

$$\Rightarrow 1 = 5(1-r) \Rightarrow r = \frac{4}{5} = 0.8$$

AG (be convinced)

A1

3

(ii) $n^{\text{th}} \text{ term} = 20 \times (0.8)^{n-1}$

Condone $20 \times (0.8)^{n-1}$

M1

$$n^{\text{th}} \text{ term} < 1 \Rightarrow 0.8^{n-1} < \frac{1}{20} \text{ oe}$$

$$0.8^{n-1} < 0.05 \text{ or } 0.8^{n-1} = k, \text{ where } k = 0.05$$

or k rounds **up** to 0.050

A1

Least n is 15

If not 15, ft on integer part of [answer (a)+2]
provided $n > 2$

A1F

SC 3/3 for 15 if no error

SC $n^{\text{th}} \text{ term} = 16^{n-1}$ M1A0A0

3

[9]

Q4.

(a) $a = -8$

B1

$$e^{2x} - 9 = 0$$

M1

$$e^{2x} = 9$$

$$2x = \ln 9$$

$$x = \ln 3$$

AG Condone verification

A1

3

(b) $(e^{2x} - 9)^2 = e^{4x} - 18e^{2x} + 81$

AG

B1

1

[4]

Q5.

(a) $y = x^{-2} \ln x$

$$\frac{dy}{dx} = x^{-2} \frac{1}{x} - 2x^{-3} \ln x$$

Use of product or quotient

M1

each term

A1 A1

$$= \frac{1 - 2 \ln x}{x^3}$$

Convincing argument $x^{-2} \times \frac{1}{x} = x^{-3}$

AG

A1

4

(b) $\int x^{-2} \ln x \, dx$ $u = \ln x$ $dv = x^{-2}$

Attempt at integration by parts

M1

$$du = \frac{1}{x} \quad v = -x^{-1}$$

A1

$$\int = -\frac{1}{x} \ln x + \int x^{-2} dx$$

A1

$$= -\frac{1}{x} \ln x - \frac{1}{x} (+C)$$

A1

4

(c) (i) At A, $\frac{dy}{dx} = 0$

$$1 - 2 \ln x = 0$$

$$\ln x = \frac{1}{2}$$

Attempt at $\ln x = k$

M1

$$x = e^{\frac{1}{2}}$$

A1

2

(ii) $R = \left[-\frac{1}{x} (\ln x + 1) \right]_1^5$
 $R = [\text{Their (b)}]_1^5$

M1

$$= -\frac{1}{5} (\ln 5 + 1) + (\ln 1 + 1)$$

OE

A1

$$= \frac{1}{5} (4 - \ln 5)$$

convincing argument; AG

A1

3

[13]

Q6.

(a) $A = 80$

B1

1

(b) $5000 = 80 \times k^{56}$

SC1 Verification. Need 62.51 or better

M1

$$k = \sqrt[56]{\frac{5000}{80}} \approx 1.07664$$

Or using logs: M1 $\ln \left(\frac{5000}{80} \right) = 56 \ln k$

$$A1 \ k = e^{\ln \left(\frac{62.5}{56} \right)}$$

Or 3/3 for $k = 1.076636$

Or 1.076637 seen

M1A1

3

(c) (i) $V = 80 \times k^{106} = 200707$

200648 using full register k

M1A1

2

(ii) $\ln 10000 = \ln k^t$

M1

$$t = \frac{\ln 10000}{\ln k} = 124.7 \Rightarrow 2024$$

M1 $t \ln k = \ln 10000$

A1 CAO

Or trial and improvement M1 expression

M1 125, 124, A1 2024

M1A1

3

[9]

Q7.

(a) (i) y -coordinate of A is $27 - 3^0 = 26$

M1A1

2

- (ii) When $x = 3, y = 27 - 3^3 = 0 \Rightarrow B(3,0)$
AG; be convinced

B1

1

- (b) $h = 1$

PI

B1

Area $\approx h/2\{\dots\}$
 $\{\dots\} = f(0) + f(3) + 2[f(1) + f(2)]$
OE summing of areas of the 'trapezia'..

M1

$\{\dots\} = "26" + 0 + 2(24 + 18)$
on (a)(i) ($\Sigma trap = "25" + 21 + 9$)

A1 ✓

(Area $\approx$) 55
on $[42 + 0.5 \times "(a)(i)"]$

A1 ✓

4

- (c) (i) $\log_{10} 3^x = \log_{10} 13$
Takes $\ln$ or $\log_{10}$ on both or $x = \log_3 13$

M1

$$x \log_{10} 3 = \log_{10} 13$$

Use of $\log 3^x = x \log 3$ or $\log_3 13 = \frac{\lg 13}{\lg 3}$ OE
(PI by $\log_3 13 = 2.335$ or better)

m1

$$x = \frac{\lg 13}{\lg 3} = 2.334717$$

$\dots = 2.3347$ to 4dp

Must show that logarithms have been used

A1

3

- (ii) $\{k =\} 14$

Condone $y = 14$; Accept final answer 14 with only zeros after decimal point e.g. 14.000

B1

1

(d) (i) Translation;

'Translation' / 'translate(d)'

B0 if more than one transformation

B1

$$\begin{bmatrix} 0 \\ -27 \end{bmatrix}$$

Accept full equivalent in words provided

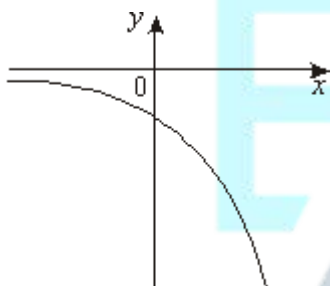
linked to 'translation/move/shift' and negative y -direction

(Note: B0 B1 is possible)

B1

2

(ii)



Correct shape (translation of given curve vertically downwards)

B1

Only point of intersection with coordinate axes is on negative y -axis and curve is asymptotic to the negative x -axis

B1

2

[15]

Q8.

(a) $y = e^{2x} - 10e^x + 12x$

(i) $\frac{dy}{dx} = 2e^{2x} - 10e^x + 12$
 $2e^{2x}$

B1

remaining terms correct, no extras

B1

2

(ii) $\frac{d^2 y}{dx^2} = 4e^{2x} - 10e^x$
ft 1 slip

B1F

1

(b) (i) $2e^{2x} - 10e^x + 12 = 0$
 $e^{2x} - 5e^x + 6 = 0$
AG (be convinced)

B1

1

(ii) $z^2 - 5z + 6 = 0$
use of $z = e^x$ oe

M1

$z = 2, 3$
 $z = 2, e^x = 2$
 $x = \ln 2$
finding $e^x = \text{their } 2, 3$

M1

$z = 3, e^x = 3$
 $x = \ln 3$
all correct AG

A1

SC: verification

$\ln 2$ (B1)

$\ln 3$ (B1)

3

(iii) $x = \ln 2$:

$y = e^{2\ln 2} - 10e^{\ln 2} + 12\ln 2$

either substitution of their $x = \ln 2$ ($e^x = 2$)

or their $x = \ln 3$ ($e^x = 3$)

M1

or
 $2^2 - 10 \times 2 + 12 \ln 2 = 4 - 20 + 12 \ln 2 = -16 + 12 \ln 2$

A1

$x = \ln 3$:

$y = e^{2 \ln 3} - 10e^{\ln 3} + 12 \ln 3 = 9 - 30 + 12 \ln 3 = -21 + 12 \ln 3$

A1

3

(iv) $x = \ln 2$:

$\frac{d^2 y}{dx^2} = 4e^{2 \ln 2} - 10e^{\ln 2}$

use of; in either of their $e^x = 2, 3$ into their $\frac{d^2 y}{dx^2}$

M1

$= 16 - 20 = -4$
 $\therefore$ maximum
 CSO

A1

$x = \ln 3$:

$\frac{d^2 y}{dx^2} = 4e^{2 \ln 3} - 10e^{\ln 3}$

$= 36 - 30 = 6$
 $\therefore$ minimum
 CSO

A1

3

[13]

Q9.

- (a) (i) $(5000 - x)$ seen in a product
Could be implied, eg $5000a - xa$

B1

$$\frac{dx}{dt} = kx(5000 - x)$$

B1

2

(ii) $200 = k \times 1000 \times (5000 - 1000)$

$$\frac{dx}{dt} = 200, x = 1000 \quad \text{in their diff. equation}$$

Condone ts and $t = 0$ for M1

M1

$$k = 0.00005$$

CAO OE

A1

2

(b) (i) $t = 4 \ln \left(\frac{4 \times 2500}{5000 - 2500} \right) = 5.5 \text{ (hours)}$

$$x \rightarrow 2500 \text{ (or } 4 \ln 4 \text{)}$$

CAO

M1

A1

2

(ii)
$$e^{\frac{30}{4}} = \frac{4x}{5000 - x}$$

(ii)

OE

B1

M1

$$5000 \times e^{7.5} = x(4 + e^{7.5})$$

Soluble for x

m1

$$x = 4988.96.. \Rightarrow 4989 \text{ rabbits infected}$$

Or 4988 or 4990; integer value only

A1

4

[10]

Q10.

- (a) {y-coordinate of A is} 2

B1

1

- (b) (i) $h = 0.25$

B1

$$\text{Integral} = \frac{h}{2} \{ \dots \}$$

$$\{ \dots \} =$$

$$f(0) + 2\left[f\left(\frac{1}{4}\right) + f\left(\frac{1}{2}\right) + f\left(\frac{3}{4}\right)\right] + f(1)$$

$$\{ \dots \} =$$

M1

$$2 + 4 + 2[(2.316.. + 2.732.. + 3.279(5..)]$$

A1 ✓

$$\{ = 6 + 2 \times 8.3276.. \} \{ = 22.65(5...) \}$$

Condone one numerical slip
Accept values to 3sf (rnd or trunc)
ft answer from (a) if not "2"

$$\text{Integral} = 0.125 \times 22.655.. = 2.8319..$$

$$\text{Integral} = 2.83 \text{ to 3 sf}$$

CAO Must be 2.83
 (NMS scores 0/4)

A1

4

- (ii) Relevant trapezia drawn on a copy of given graph

Accept relevant single trapezium with its
sloping side above the curve

M1

{Approximation is an}overestimate

A1

2

(c) $5 = 3^x + 1 \Rightarrow 3^x = 4$

B1

$$\log_{10} 3^x = \log_{10} 4$$

Takes $\ln$ or $\log_{10}$ on both sides of $3^x = k$, where $k > 0$

M1

$$x \log_{10} 3 = \log_{10} 4$$

Use of $\log 3^x = x \log 3$

m1

$$x = \frac{\lg 4}{\lg 3} = 1.26185 \dots$$

$= 1.2619$ to 4dp

*Accept 4dp or better
[If using T&I a full justification is
required; else M0m0A0]*

A1

4

(d) $f(x) = 3^{-x} + 1$

B1

1

[12]

Q11.

(a) $2e^x = 5$

$$e^x = \frac{5}{2}$$

(exact)

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M1

$$x = \ln \frac{5}{2} \quad (0.916)$$

(A0 if further wrong work)

A1

2

(b) (i) $2e^x + 5e^{-x} = 7$

$$2e^{2x} + 5 = 7e^x$$

Dealing with e^{-x}

M1

$$2y^2 - 7y + 5 = 0$$

AG

A1

2

(ii) $(2y - 5)(y - 1) = 0$

attempt to solve $y = \frac{5}{2}, 1$ (SC B1)

M1

$$x = \ln \frac{5}{2}$$

$$e^x = \frac{5}{2}$$

A1

$$x = 0 \text{ (or } \ln 1)$$

$$e^x = 1$$

A1

3

[7]

Q12.

(a) (i) $t = 0$

$$x = 85$$

B1

1

(ii) $t = 30$

$$x = 15 + 70e^{-\frac{30}{40}}$$

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M1

$$= 48.06 \approx 48^\circ\text{C}$$

A1

2

(iii) $x = 60$

$$\frac{60 - 15}{70} = e^{-\frac{t}{40}}$$

M1

$$\ln \left(\frac{45}{70} \right) = -\frac{t}{40}$$

Their 60 - 15

m1

$$t = 17.67 \approx 18 \text{ minutes}$$

A1

3

(b) (i) $\int \frac{dx}{x-15} = -\int \frac{dt}{40}$

Attempt to separate and integrate

M1

Correct expression and used

A1

$$\ln(x-15) = -\frac{t}{40} (+c)$$

A1A1

$$(0, 85) \quad c = \ln 70$$

m1

$$\frac{t}{40} = \ln 70 - \ln(x-15) \quad t = 40 \ln \left(\frac{70}{x-15} \right)$$

*Use (0.85) to find c, which must now appear in expression
Manipulate to t= ...*

A1

6

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(ii) $e^{\frac{t}{40}} = \frac{70}{x-15}$

Manipulate expression including a ln towards x = ...

M1

$$x-15 = 70e^{-\frac{t}{40}}$$

AG; convincingly obtained

M1A1

2

[14]